

# RATIONAL METHOD HYDROLOGY STUDY

2026 County of San Diego Hydrology Manual, Sections 3 and 6

| PROJECT          |                                              |
|------------------|----------------------------------------------|
| Project          | La Jolla Drainage Study                      |
| Location         | La Jolla, California, USA                    |
| Engineer         | Kiran Pallachulla, PE                        |
| Firm             | Snipes-Dye Associates                        |
| Date             | 2026-09-15                                   |
| Jurisdiction     | SD County 2026     Storm frequency: 100-year |
| Site coordinates | 32.81230, -117.26770                         |

| NOAA ATLAS 14 DEPTHS (INCHES) |        |        |        |
|-------------------------------|--------|--------|--------|
| 5-min                         | 10-min | 15-min | 30-min |
| 0.329                         | 0.472  | 0.570  | 0.791  |
| 60-min                        | 2-hr   | 3-hr   | 6-hr   |
| 1.120                         | 1.490  | 1.770  | 2.360  |

| SUBAREA SUMMARY |           |       |      |      |      |           |          |
|-----------------|-----------|-------|------|------|------|-----------|----------|
| Subarea         | Area (ac) | C     | Ti   | Tt   | Tc   | I (in/hr) | Qp (cfs) |
| Subarea A       | 0.44      | 0.510 | 7.40 | 1.22 | 8.60 | 3.044     | 0.68     |
| Subarea B       | 0.08      | 0.510 | 7.40 | 1.22 | 8.60 | 3.044     | 0.12     |
| Subarea C       | 0.45      | 0.510 | 7.40 | 1.22 | 8.60 | 3.044     | 0.70     |
| TOTAL           | 0.97      |       |      |      |      |           | 1.50     |

| FLOW PATH SUMMARY |               |        |       |        |         |          |
|-------------------|---------------|--------|-------|--------|---------|----------|
| Flow path         | Type          | L (ft) | S (%) | n      | V (fps) | Tt (min) |
| Flow path 1       | Street gutter | 284    | 1.979 | 0.0130 | 5.51    | 0.86     |
| Flow path 2       | Swale / ditch | 197    | 0.604 | 0.0300 | 1.73    | 1.90     |
| Flow path 4       | Street gutter | 295    | 2.010 | 0.0130 | 5.56    | 0.88     |
| Flow path 4       | Street gutter | 115    | 3.183 | 0.0130 | 6.99    | 0.27     |

## NODE AND STATION LISTING

SAN DIEGO COUNTY RATIONAL HYDROLOGY  
2026 County of San Diego Hydrology Manual, Sections 3 and 6  
HydroRational Rational Hydrology Study

PROJECT : La Jolla Drainage Study  
LOCATION : La Jolla, California, USA  
ENGINEER : Kiran Pallachulla, PE  
FIRM : Snipes-Dye Associates  
DATE : 2026-09-15  
JURISDICTION : SD County 2026 STORM FREQUENCY: 100 YEAR  
SITE COORDS : 32.81230, -117.26770

### \*\*\*\*\* HYDROLOGY STUDY CONTROL INFORMATION \*\*\*\*\*

ENGLISH (IN-LB) UNITS USED  
RATIONAL METHOD PER SECTION 3,  $Q = C I A$   
INTENSITY BY LOG-LOG INTERPOLATION OF NOAA ATLAS 14 DEPTHS, SECTION 3.1.3  
INITIAL TIME OF CONCENTRATION PER TABLE 3-2 (FAA OVERLAND FLOW)  
RUNOFF COEFFICIENT, TABLE 3-1 AND SECTION 3.1.2:  
 $C = 0.90 \times (\% \text{ IMPERVIOUS}) + C_p \times (1 - \% \text{ IMPERVIOUS})$   
0.90 IS THE RUNOFF COEFFICIENT OF IMPERVIOUS SURFACE  
 $C_p$  IS THE PERVIOUS COEFFICIENT BY SOIL GROUP:  
 $A = 0.20$ ,  $B = 0.25$ ,  $C = 0.30$ ,  $D = 0.35$

### \*\*\*\*\* NOAA ATLAS 14 POINT PRECIPITATION DEPTHS (INCHES) \*\*\*\*\*

5-MIN = 0.329    10-MIN = 0.472    15-MIN = 0.570    30-MIN = 0.791  
60-MIN = 1.120    2-HR = 1.490    3-HR = 1.770    6-HR = 2.360

+++++  
FLOW PROCESS FROM NODE 1.00 TO NODE 2.00 IS CODE = 21  
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### >>>>RATIONAL METHOD INITIAL SUBAREA ANALYSIS<<<<

\*USER SPECIFIED(SUBAREA):  
SUBAREA IDENTIFICATION = Subarea A  
LAND USE (TABLE 3-1) = MDR-7.3 SOIL GROUP = B  
PERCENT IMPERVIOUS = 40.0  
IMPERVIOUS AREA FRACTION = .4000  
USER-SPECIFIED RUNOFF COEFFICIENT = .5100  
INITIAL SUBAREA FLOW-LENGTH(FEET) = 70.00  
DOWNSTREAM ELEVATION(FEET) = 84.84  
SUBAREA OVERLAND FLOW SLOPE(PERCENT) = 2.00  
SUBAREA OVERLAND TIME OF FLOW(MIN.) = 7.400  
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044  
SUBAREA AREA(ACRES) = 0.44  
SUBAREA RUNOFF(CFS) = 0.68  
TOTAL AREA(ACRES) = 0.44 TOTAL RUNOFF(CFS) = 0.68  
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SUBAREA TIME OF CONCENTRATION TC(MIN.) = 8.600 (manual entry)  
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044  
PEAK FLOW RATE(CFS) AT THIS POINT = 0.68  
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FLOW PROCESS FROM NODE 2.00 TO NODE 3.00 IS CODE = 21  
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### >>>>RATIONAL METHOD INITIAL SUBAREA ANALYSIS<<<<

\*USER SPECIFIED(SUBAREA):  
SUBAREA IDENTIFICATION = Subarea B  
LAND USE (TABLE 3-1) = MDR-7.3 SOIL GROUP = B  
PERCENT IMPERVIOUS = 40.0  
IMPERVIOUS AREA FRACTION = .4000  
USER-SPECIFIED RUNOFF COEFFICIENT = .5100  
INITIAL SUBAREA FLOW-LENGTH(FEET) = 70.00  
DOWNSTREAM ELEVATION(FEET) = 84.84  
SUBAREA OVERLAND FLOW SLOPE(PERCENT) = 2.00  
SUBAREA OVERLAND TIME OF FLOW(MIN.) = 7.400  
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044  
SUBAREA AREA(ACRES) = 0.08  
SUBAREA RUNOFF(CFS) = 0.12  
TOTAL AREA(ACRES) = 0.08 TOTAL RUNOFF(CFS) = 0.12  
-----

SUBAREA TIME OF CONCENTRATION TC(MIN.) = 8.600 (manual entry)  
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044  
PEAK FLOW RATE(CFS) AT THIS POINT = 0.12  
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FLOW PROCESS FROM NODE 3.00 TO NODE 4.00 IS CODE = 21  
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### >>>>RATIONAL METHOD INITIAL SUBAREA ANALYSIS<<<<

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*USER SPECIFIED(SUBAREA):
SUBAREA IDENTIFICATION          = Subarea C
LAND USE (TABLE 3-1)           = MDR-7.3   SOIL GROUP = B
PERCENT IMPERVIOUS              = 40.0
IMPERVIOUS AREA FRACTION        = .4000
USER-SPECIFIED RUNOFF COEFFICIENT = .5100
INITIAL SUBAREA FLOW-LENGTH(FEET) = 70.00
DOWNSTREAM ELEVATION(FEET)      = 84.84
SUBAREA OVERLAND FLOW SLOPE(PERCENT) = 2.00
SUBAREA OVERLAND TIME OF FLOW(MIN.) = 7.400
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044
SUBAREA AREA(ACRES)             = 0.45
SUBAREA RUNOFF(CFS)             = 0.70
TOTAL AREA(ACRES) = 0.45  TOTAL RUNOFF(CFS) = 0.70
-----
SUBAREA TIME OF CONCENTRATION TC(MIN.) = 8.600 (manual entry)
100 YEAR RAINFALL INTENSITY(INCH/HOUR) = 3.044
PEAK FLOW RATE(CFS) AT THIS POINT = 0.70
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FLOW PROCESS AT NODE      Node 4          IS CODE = 1
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>>>>>CONFLUENCE OF MINOR STREAMS<<<<<
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** CONFLUENCE DATA **
STREAM SOURCE      RUNOFF      TC      INTENSITY      AREA
NUMBER              (CFS)      (MIN.)  (INCH/HR)  (ACRES)
1  Subarea A        0.68      8.60      3.044      0.44
2  Subarea B        0.12      8.60      3.044      0.08
3  Subarea C        0.70      8.60      3.044      0.45
RULE APPLIED: Note 2, Tc values within 10 percent
CONTROLLING TIME OF CONCENTRATION(MIN.) = 8.60
TOTAL AREA(ACRES) = 0.97  PEAK FLOW RATE(CFS) = 1.51
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SUBAREA SUMMARY TABLE
-----
SUBAREA      AREA      C      Ti      Tt      Tc      I      Q
              (AC)              (MIN) (MIN) (MIN) (IN/HR) (CFS)
-----
Subarea A    0.44    .5100  7.40  1.22  8.60  3.044  0.68
Subarea B    0.08    .5100  7.40  1.22  8.60  3.044  0.12
Subarea C    0.45    .5100  7.40  1.22  8.60  3.044  0.70
-----
TOTAL        0.97                                1.50
-----
NODE SUMMARY TABLE
-----
NODE      TYPE      ELEV(FT)  STREAMS  AREA(AC)  Tc(MIN)  Q(CFS)
-----
Node 1    JUNCTION    86.80      0        -        -        -
Node 2    OUTFALL      81.18      1        0.97     8.87     1.49
Node 3    JUNCTION    86.03      0        -        -        -
Node 4    JUNCTION    84.84      3        0.97     8.60     1.51
Node 5    JUNCTION    90.77      0        -        -        -
-----
FLOW PATH SUMMARY TABLE
-----
FLOW PATH  TYPE      L(FT)  S(%)  n      V(FPS)  Tt(MIN)
-----
Flow path 1  Street gutter  284    1.979  0.0130  5.51    0.86
Flow path 2  Swale / ditch  197    0.604  0.0300  1.73    1.90
Flow path 4  Street gutter  295    2.010  0.0130  5.56    0.88
Flow path 4  Street gutter  115    3.183  0.0130  6.99    0.27
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SECTION 6 RATIONAL METHOD HYDROGRAPH
-----
STORM DURATION          = 6 HOURS
COMPUTATION INTERVAL Tc(MIN.) = 9
NUMBER OF RAINFALL BLOCKS N = 40
PEAK FLOW Qp(CFS)       = 0.68
TIME TO PEAK Tp(MIN.)   = 245 (240 + 0.5 x Tc)
RUNOFF VOLUME(AC-IN)    = 0.53
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END OF RATIONAL METHOD ANALYSIS
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Generated by HydroRational. Results are the responsibility of the
engineer of record and should be checked against the 2026 County of
San Diego Hydrology Manual before submittal.

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